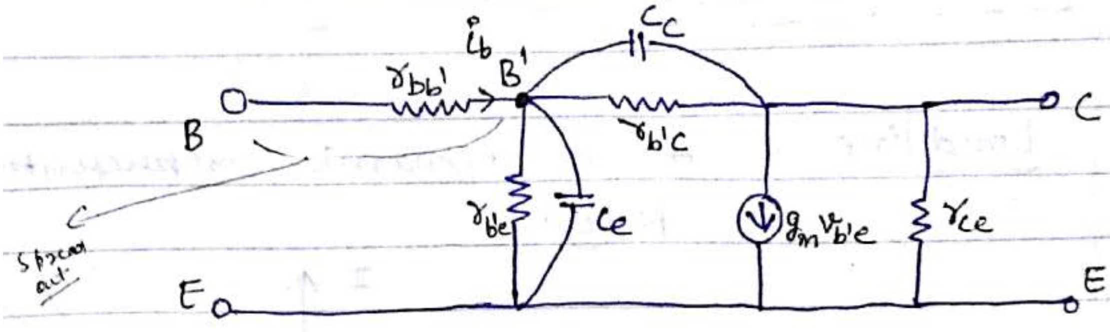
[High frequency analysis of BJT] :-

This model proposed by Gia colletto —

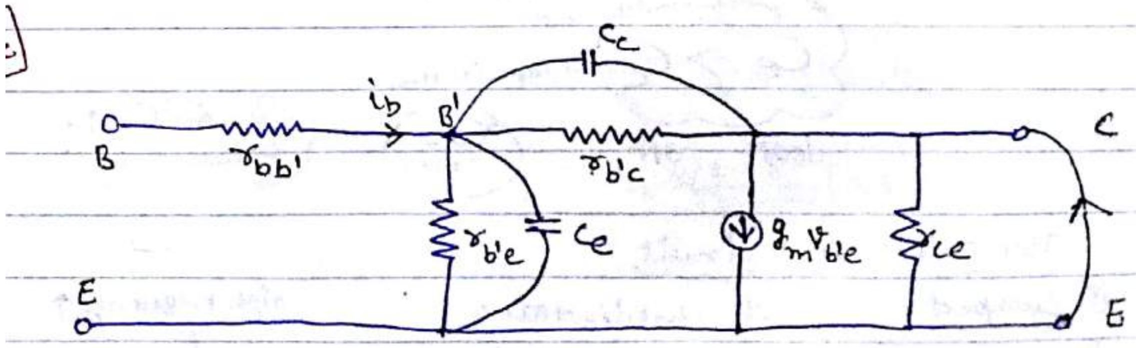


Due to high frequency, Now circuit will become $\boxed{\text{reactive}}$.

X_c reactance value mF & μF

* All external capacitance are short circuited and all internal capacitance are open circuited

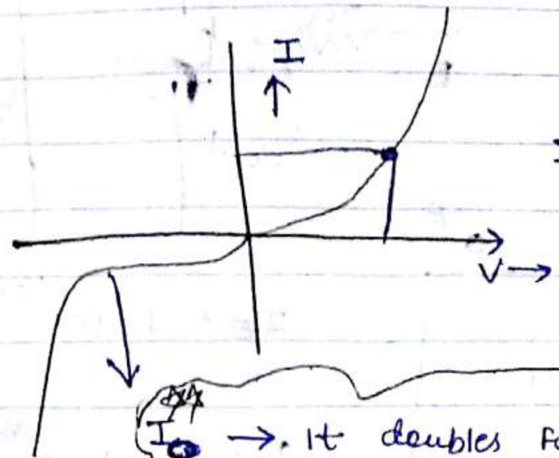
value in pF so we treat as open circuited



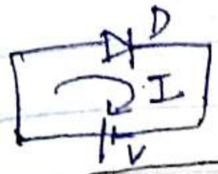
$$g_m = \frac{i_c}{v_{b'e}}\bigg|_{V_{CE}=0}$$

g_m → Short circuit trans conductance gain.

$$g_m = \frac{\partial i_c}{\partial v_{B'E}}\bigg|_{v_{ce}=0}$$



$I_Q \rightarrow$ Qscient current



$I_o \rightarrow$ It doubles for every increase in 10°c.

$$I = I_o(e^{V/V_T} - 1)$$ ← Diode equatn

$V_T = \frac{KT}{q}$; 26mV at 27°c

$V_T \rightarrow$ Thermal voltage

$I_o \rightarrow$ Reverse saturatn current

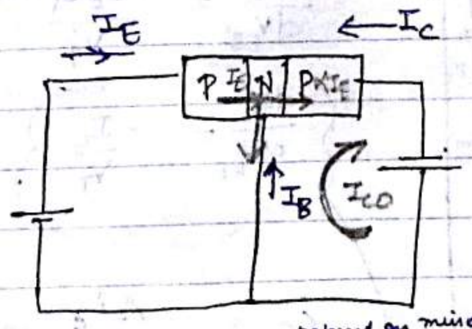
Forward : $I = I_o e^{V/V_T}$

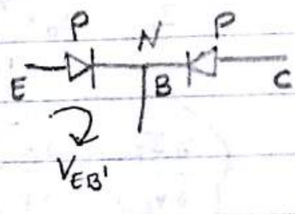
Reverse : $I \approx -I_o$.

$$\frac{dI}{dV} = \frac{I_o e^{V/V_T}}{V_T} \approx \frac{I_Q}{V_T}$$

dynamic resistance

$$r = \frac{V_T}{I_Q}$$





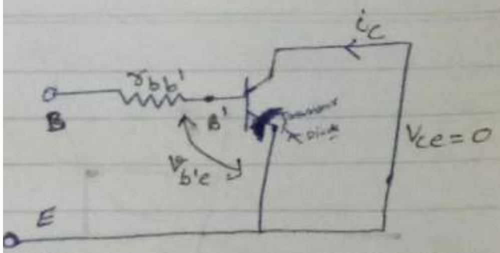
$i_C = I_{CO} - \alpha i_E$

$I_C = I_{CO} - \alpha I_E$ ← Transistor equatn

I_{CO} → Depend on minority charge carrier & temp.

I_{CO} → reverse saturatn current in Base collector junctn.

always +ve either in PNP or NPN



$$I_E = I_o\left(e^{V_{EB'}/V_T} - 1\right)$$

$$\frac{dI_E}{dV_{EB'}} = \frac{I_o e^{V_{EB'}/V_T}}{V_T} \approx \frac{I_E}{V_T}$$

$$r_E = \frac{V_T}{I_E}$$

$$\frac{dI_c}{dV_{EB'}} = I_{co} - \alpha I_E$$

$$\frac{dI_c}{dV_{EB'}} = 0 - \frac{\alpha}{r_E}$$

$$\frac{dI_c}{dV_{EB'}} = \frac{-I_E\alpha}{V_T}$$

$$\frac{dI_c}{dV_{B'E}} = -\frac{I_c}{V_T}$$

$$i_c = \Delta i_C$$

$$i_C = I_{CQ} + i_c$$

$$\frac{i_c}{V_{b'e}} = \frac{-I_c}{V_T}$$

$$g_m = \frac{-I_{CQ}}{V_T}$$

Que- For a NPN transistor the Q-sient collector 1.3 mA. If the transistor is operated at room temp Calculate the short circuit the trans conductance gain of such a conductor. If suddenly the Q-point of the transistor is shifted & now the collector current (Q-current) is maintain 10-times of the earlier New What will be Transconductance gain? what do you conclude from these 2 options?

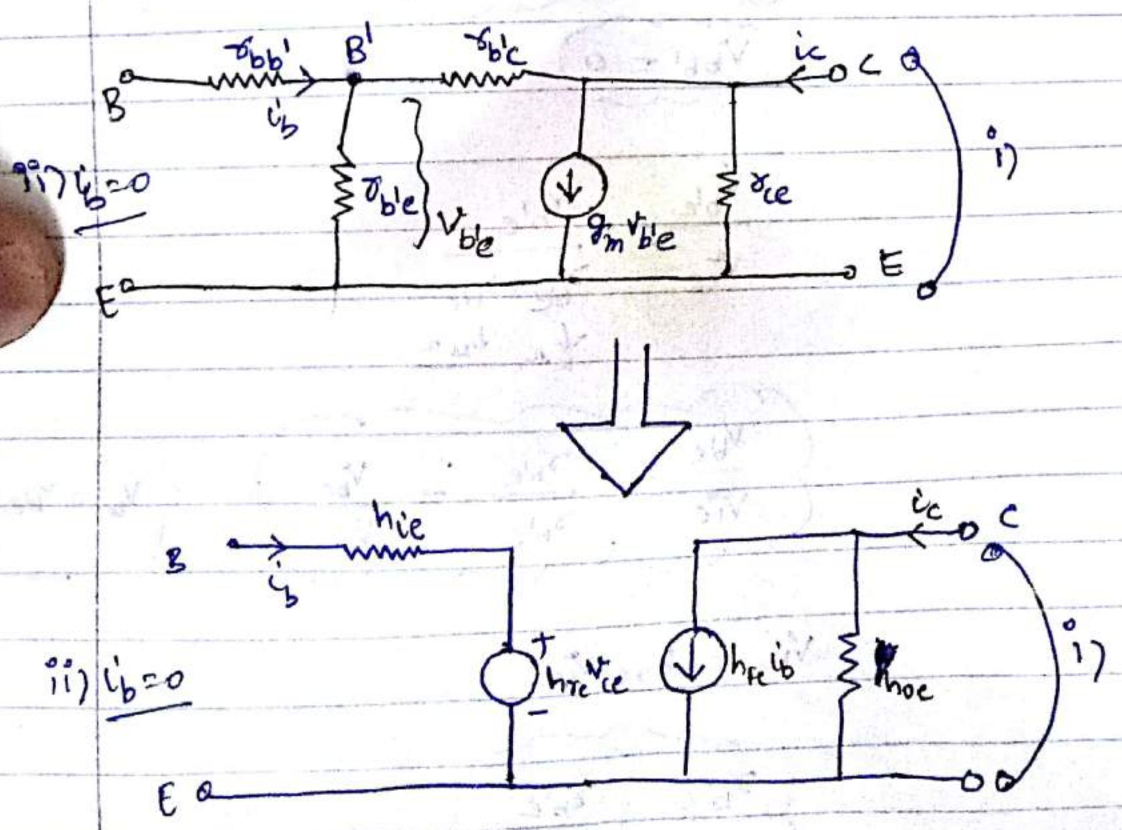
Sol:-

i) $g_m = \frac{I_{CQ}}{V_T} = .05$ or $\simeq$ 50mA/V

ii) $g_m = 0.5 \simeq$ 500 mA/V

A — BJT Transistor provides us a trans-conductance gain of 500 mA/V but FET unable to provide 500 mA/V it only provide us with difficulty 50mA/V

HIGH → LOW



i) $i_c = h_{fe} i_b$

$i_c = g_m v_{b'e}$

or $g_m v_{b'e} = h_{fe} i_b$

$$\left.\frac{v_{b'e}}{i_b} = \frac{h_{fe}}{g_m}\right|_{v_{ce}=0}$$

Now $r_{b'c} \rightarrow$ it is very large compare to $r_{b'e}$.

base collector reverse bias junction dynamic resistance

$\downarrow$ MΩ.

So;

$$r_{b'e} = \left.\frac{v_{b'e}}{i_b}\right|_{v_{ce}=0}$$

$$r_{b'e} = \frac{h_{fe}}{g_m} \approx 1K\Omega$$ So simply

ii) $i_b = 0$ then

$V_{bb'} = 0$

$$\frac{v_{b'e}}{v_{ce}} = \frac{r_{b'e}}{\underbrace{r_{b'e}}_{K\Omega} + \underbrace{r_{b'c}}_{M\Omega}}$$

$$\frac{v_{b'e}}{v_{ce}} = \frac{r_{b'e}}{r_{b'c}} = \frac{v_{be}}{v_{ce}}$$ $\{v_{be} = v_{b'e}\}$

$v_{be} = h_{re} v_{ce}$.

$$r_{b'c} = \frac{r_{b'e}}{h_{re}} \approx 4M\Omega$$

iii) $i_c = V_{ce}\, g_{ce} + \underline{g_m v_{b'e}} + g_{b'c} v_{ce} \Big|_{i_b = 0}$ —— (I)

$i_c = h_{oe} V_{ce} \Big|_{i_b = 0}$ —— (II)

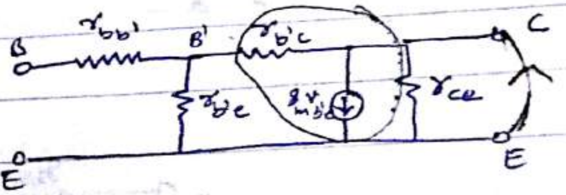
$h_{oe} V_{ce} = V_{ce}\, g_{ce} + g_m V_{be} + g_{b'c} V_{ce} \Big|_{i_b = 0}$

$[V_{be} = h_{re} V_{ce}]$

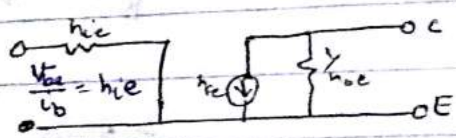
$h_{oe} \cancel{V_{ce}} = \frac{\cancel{V_{ce}}}{r_{ce}} + g_m h_{re} \cancel{V_{ce}} + \frac{\cancel{V_{ce}}}{r_{b'c}}$

$g_{ce} = h_{oe} - g_m h_{re} - g_{b'c}$ $\quad$ $r_{ce} \simeq 80K\Omega$

iv) $\underline{V_{ce} = 0}$

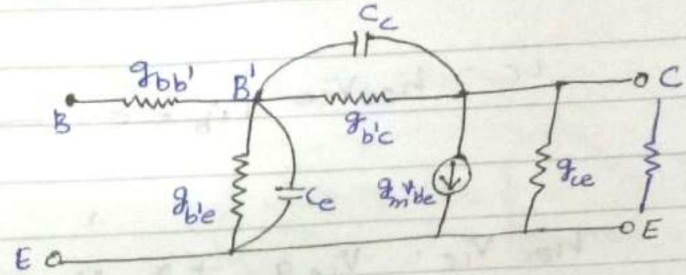


$\frac{V_{be}}{i_b} = r_{bb'} + r_{b'e}$



$h_{ie} - r_{b'e} = r_{bb'}$ $\quad$ $\approx 100\Omega$

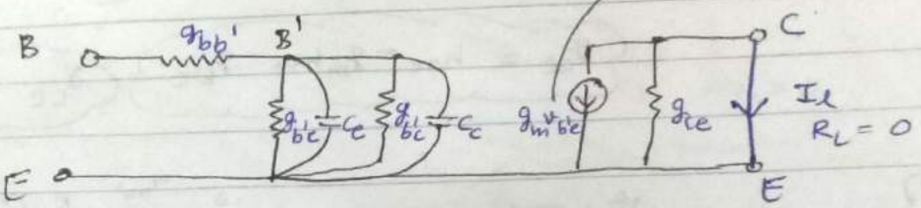
S.C. current gain of BJT at high Frequency :-

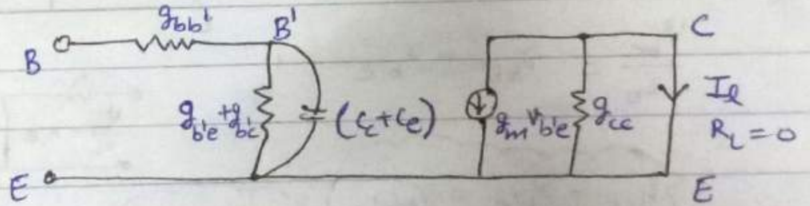


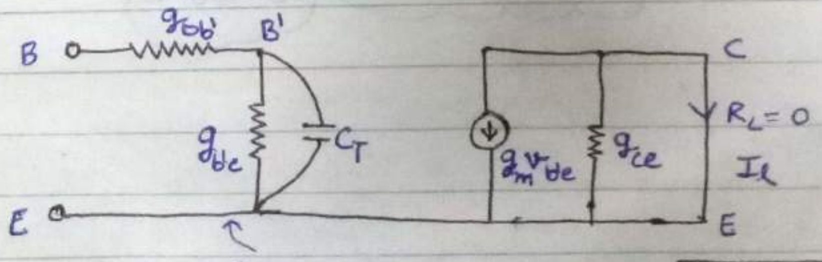
$(G_L = \infty)$ (For s.c) i.e $(R_L = 0)$

$\underline{g_{bb'} >> g_{b'e}}$

$v_{b'e}$ RMS (for analysing RLC ckt)







$\underline{g_{b'e} >> g_{b'c}}$

$\boxed{C_T = C_e + C_c}$

$$A_i = \frac{I_L}{I_i}\Bigg|_{v_{ce}=0\,;\; R_L = 0}$$

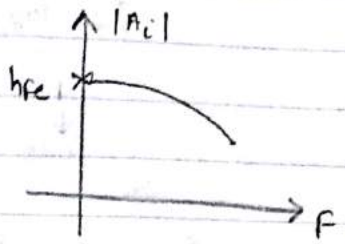
$$= \frac{-g_m v_{b'e} \rightarrow \{v_{b'e}\}_{RMS}}{Y \cdot v_{b'e}} \qquad \{Y = g_{b'e} + j\omega C_T\}$$

$I_b = Y\, v_{b'e}$ (rms, rms)

$$= \frac{-g_m}{g_{b'e} + j\omega C_T} \leftarrow \text{Remember}$$

or $$A_i = \frac{-\dfrac{g_m}{g_{b'e}}}{1 + \dfrac{j\omega C_T}{g_{b'e}}}$$

or $$A_i = \frac{-h_{fe}}{1 + \dfrac{j\omega C_T}{g_{b'e}}}$$



$$|A_i| = \frac{h_{fe}}{\sqrt{1 + (2\pi f \, r_{b'e} C_T)^2}}$$

$$|A_i|_{max} = h_{fe}$$

Now,

$$A_i = \frac{-h_{fe}}{1 + j\left(\frac{f}{f_\beta}\right)}$$

$$f_\beta = \frac{1}{2\pi r_{b'e} C_T}$$

f_β is
- 3 dB freq.
- cut off freq.
- Half power point freq.
- band width freq.

→ If h_{fe} is taken to be const

$$|A_i| = \frac{h_{fe}}{\sqrt{1 + \left(\frac{f}{f_\beta}\right)^2}}$$

* [Power get reduced to half at $f = f_\beta$; i.e if we ↑ the freq. upto f_β then power get reduced to half. Hence it is called half power point frequency]

$f = 0$; $|A_i| = h_{fe}$

|

|

$f = f_\beta$; $|A_i| = \frac{h_{fe}}{\sqrt{2}}$

* [.707 times of max. is cut-off frequency → f_β]
* Amplifier is only working b/w $[0 - f_m]$.

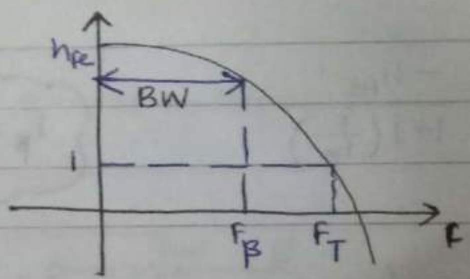
* Amplifier gets cutoff beyond the frequency F_m according to IEEE standards.

* BW gets zero beyond the half power frequency

$$\boxed{20 \log_{10} |A_i| \, dB.}$$

$F \rightarrow 0dB$ $\quad F_\beta \rightarrow 3dB$ [If frequency has gone to max. from 3dB then we reject the amplifier].

—: [F_T is called unity gain Transition Frequency]



$$1 = \frac{h_{fe}}{\sqrt{1+\left(\frac{F_T}{F_\beta}\right)^2}}$$

$$1 + \left(\frac{F_T}{F_\beta}\right)^2 = h_{fe}^2$$

$$\frac{F_T}{F_\beta} = \sqrt{h_{fe}^2 - 1}$$

$$\frac{F_T}{F_\beta} = h_{fe}$$

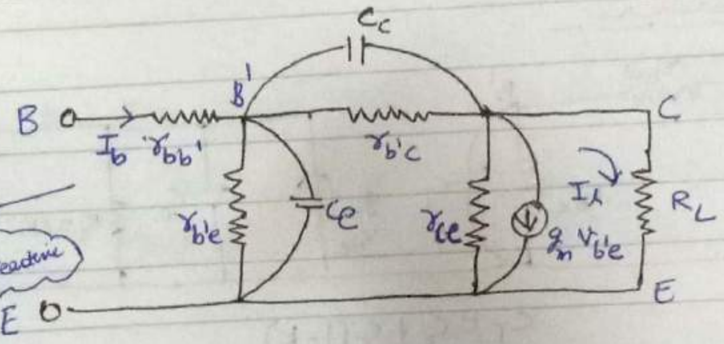
$$\boxed{F_T = h_{fe} \times F_\beta}$$

$\boxed{F_T = \underbrace{h_{fe}}_{\text{Gain}} \times \underbrace{F_\beta}_{BW}}$ = Gain BW product

* Gain BW product is always Const.

—x —x —x —x —x —x

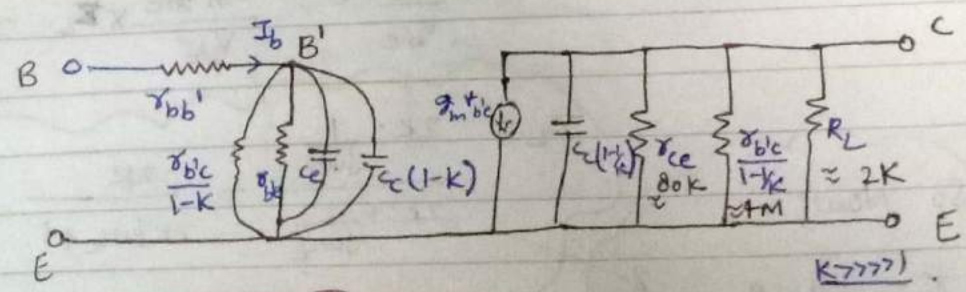
To Find current Gain of same BJT in presence of a Load :—



RMS value for complex algebra of a reactive ckt

$$A_I = \frac{I_L}{I_b} \qquad K = \frac{V_{ce}}{V_{b'e}}$$

By Millman's Theorem :—



$K >>>> 1$

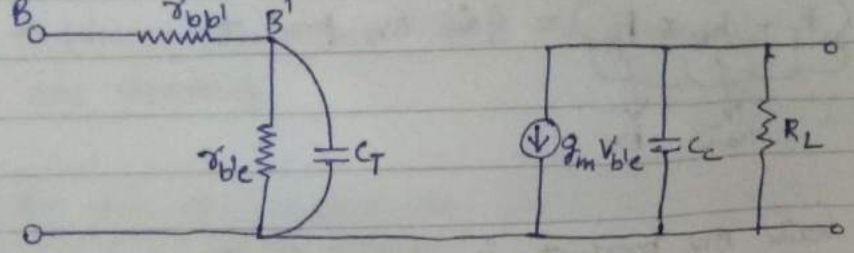
$C_c\left(1-\frac{1}{K}\right) = C_c$

$r_{b'c}\left(1-\frac{1}{K}\right) = r_{b'c}$

$\boxed{\text{Wide band Amplifier } R_L \le 2K}$

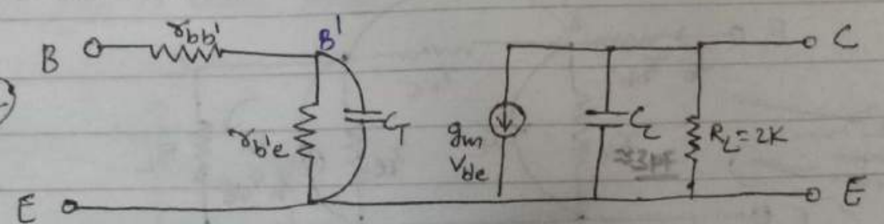
Now R_L is dominate over $r_{b'c}$ & r_{ce}.

& $r_{b'e}$ is dominate over $\frac{r_{b'c}}{1-K}$



$$\left[C_T = C_e + C_c(1-K)\right]$$

Time constant :-



(f = 10 MHz)

There are 2 poles in ckt bcz there are 2 discharging ckt in the diagram but we have to simplify into single one

$$C_T = C_e + C_c(1-K)$$

$$\tau_o = 2K \times 3pF \quad \text{o/p}$$
$$\tau_o = 6ns$$

$$K = \frac{V_{ce}}{V_{b'e}} = \frac{-g_m \cancel{V_{b'e}}}{\cancel{V_{b'e}}} \times Z_L \approx -g_m Z_L$$

$$Z_L = \frac{2K \cdot \frac{1}{j\omega C_c}}{2K + \frac{1}{j\omega C_c}} = \frac{2K}{2Kj\omega C_c + 1} = \frac{2K}{\sqrt{1 + 4K^2\omega^2 C_c^2}}$$

$$= \frac{2K}{\sqrt{1 + 4 \times 10^6 \times 4\pi^2 \times 10^{14} \times 9 \times 10^{-24}}}$$

$$= \frac{2K}{\sqrt{1.14}}$$

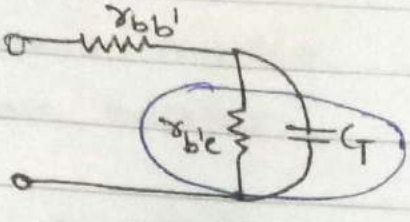
$$\approx 2K = R_L$$

So Now;

$$K = -g_m R_L$$

$$\& \; C_T = C_e + C_c(1-K)$$

$$C_T = C_e + C_c(1+g_m R_L)$$
$$= 100 + 3(1+100)$$
$$C_T \cong 403\,pF$$



$$\tau = 403pF \times 1000$$
$$\tau_i = 403\,ns$$

* In these 2 time const ; bigger one (τ_i) is dominant over smaller one (τ_o) because it take more time to stable.

$$f_\beta = \frac{1}{2\pi RC}$$

$$A_I = \frac{I_L}{I_b} = \frac{-g_m V_{b'e}}{V_{b'e}[g_{b'e} + j\omega C_T]}$$

$$A_I = \frac{-h_{fe}}{\left[1 + j\frac{f}{f_H}\right]}$$

where $f_H = \frac{1}{2\pi r_{b'e}(C_e + C_c(1+g_m R_L))}$ *